

Cascaded $\chi^{(2)}: \chi^{(2)} \rightarrow \chi_{\text{eff}}^{(3)}$ in a Doubly-Resonant Metasurface

Nikita Nefedkin

May 27, 2026

1 Conventions

Real fields written as sums of complex envelopes,

$$\mathcal{E}_i(\mathbf{r}, t) = \frac{1}{2} \sum_n [E_{n,i}(\mathbf{r}, t) e^{-i\omega_n t} + \text{c.c.}], \quad \mathcal{P}_i(\mathbf{r}, t) = \frac{1}{2} \sum_n [P_{n,i}(\mathbf{r}, t) e^{-i\omega_n t} + \text{c.c.}], \quad (1)$$

with $E_{n,i}, P_{n,i}$ slowly varying. The second-order polarisation

$$P_i^{(2)}(\omega_\sigma) = \varepsilon_0 \sum_{(p,q)} \chi_{ijk}^{(2)}(\omega_\sigma; \omega_p, \omega_q) E_j(\omega_p) E_k(\omega_q), \quad (2)$$

gives for the two relevant channels

$$P_i^{(2)}(2\omega_1) = \varepsilon_0 \chi_{ijk}^{(2)}(2\omega_1; \omega_1, \omega_1) E_j(\omega_1) E_k(\omega_1) \quad (\text{SHG}) \quad (3)$$

$$P_i^{(2)}(\omega_1) = 2\varepsilon_0 \chi_{ijk}^{(2)}(\omega_1; 2\omega_1, -\omega_1) E_j(2\omega_1) E_k^*(\omega_1) \quad (\text{DFG}) \quad (4)$$

The factor 2 counts orderings $(2\omega_1, -\omega_1)$ and $(-\omega_1, 2\omega_1)$. The native Kerr response is

$$P_i^{(3)}(\omega_1) = 3\varepsilon_0 \chi_{ijkl}^{(3)}(\omega_1; \omega_1, \omega_1, -\omega_1) E_j(\omega_1) E_k(\omega_1) E_l^*(\omega_1). \quad (5)$$

Under Kleinman symmetry

$$\chi_{ijk}^{(2)}(2\omega_1; \omega_1, \omega_1) = \chi_{kij}^{(2)}(\omega_1; 2\omega_1, -\omega_1) \equiv \chi_{ijk}^{(2)}(\mathbf{r}). \quad (6)$$

Scalar $\chi^{(2)}$. The symbol $\chi_{ijk}^{(2)}(\mathbf{r})$ *with* indices is always the full position-dependent rank-3 tensor introduced above. The *unsubscripted* symbol $\chi^{(2)}$ (and its modulus $|\chi^{(2)}|$) is a single scalar magnitude characterising the active component of that tensor inside the nonlinear region $V_{\text{NL}} \subseteq V_{\text{eff}}$:

$$|\chi^{(2)}| \equiv \max_{(ijk), \mathbf{r} \in V_{\text{NL}}} |\chi_{ijk}^{(2)}(\mathbf{r})|. \quad (7)$$

For a homogeneous host with a single dominant tensor element, $|\chi^{(2)}|$ is just that element (e.g. $|\chi^{(2)}| = 2d_{33}$ in LiNbO₃ with the z -polarised eigenmodes; $|\chi^{(2)}| = 2d_{14}$ in GaAs/GaP/AlGaAs; etc.). In the bulk SVEA expressions of §3(a), $\chi^{(2)}$ denotes the same scalar projected onto the plane-wave polarisations, taken real and positive by an overall choice of phase.

This scalar is the dimensional prefactor that we pull out of the tensorial integrals to make the geometric coefficients $\tilde{\eta}$ and η_{geom} in §5 dimensionless. With this convention, $|\chi^{(2)}|^2$ in (15) and (34) carries the material units, and all mode-shape, fill-fraction, and tensor-projection information sits in η_{geom} with $0 \leq \eta_{\text{geom}} \leq 1$.

2 Modes and energy normalisation

Linear eigenmodes $\mathbf{f}_n(\mathbf{r})$, $\mathbf{h}_n(\mathbf{r})$ satisfy

$$\nabla \times \nabla \times \mathbf{f}_n(\mathbf{r}) = \frac{\omega_n^2}{c^2} \varepsilon_r(\mathbf{r}, \omega_n) \mathbf{f}_n(\mathbf{r}), \quad (8)$$

with $\mathbf{h}_n = -i\nabla \times \mathbf{f}_n / (\omega_n \mu_0)$. Brillouin energy normalisation:

$$\frac{1}{2} \int [\varepsilon_0 \tilde{\varepsilon}_n(\mathbf{r}) |\mathbf{f}_n|^2 + \mu_0 |\mathbf{h}_n|^2] d^3r = 1, \quad \tilde{\varepsilon}_n(\mathbf{r}) \equiv \left. \frac{\partial(\omega \varepsilon_r(\mathbf{r}, \omega))}{\partial \omega} \right|_{\omega=\omega_n}. \quad (9)$$

Modal amplitude $a_n(t)$ defined by $E_{n,i}(\mathbf{r}, t) = a_n(t) f_{n,i}(\mathbf{r})$; time-averaged energy in mode n is $|a_n|^2$.

3 Cascaded polarisation from $\chi^{(2)} \cdot G \cdot \chi^{(2)}$

The SH field at leading order solves

$$[\nabla \times \nabla \times - k_2^2 \varepsilon_r(\mathbf{r}, 2\omega_1)] \mathbf{E}(2\omega_1) = k_2^2 \mu_0 c^2 \mathbf{P}^{\text{NL}}(2\omega_1), \quad k_2 \equiv 2\omega_1/c, \quad (10)$$

with dyadic Green's function $\bar{\bar{G}}(\mathbf{r}, \mathbf{r}'; 2\omega_1)$:

$$E_j(\mathbf{r}, 2\omega_1) = k_2^2 \int d^3r' G_{jl}(\mathbf{r}, \mathbf{r}'; 2\omega_1) \chi_{lmn}^{(2)}(\mathbf{r}') E_m(\mathbf{r}', \omega_1) E_n(\mathbf{r}', \omega_1). \quad (11)$$

Substituting into the DFG polarisation (4),

$$P_i^{\text{casc}}(\mathbf{r}, \omega_1) = 2\varepsilon_0 k_2^2 \int d^3r' \underbrace{\chi_{ijk}^{(2)}(\mathbf{r}) G_{jl}(\mathbf{r}, \mathbf{r}'; 2\omega_1) \chi_{lmn}^{(2)}(\mathbf{r}')}_{\equiv \Xi_{ikmn}^{\text{casc}}(\mathbf{r}, \mathbf{r}')} E_m(\mathbf{r}', \omega_1) E_n(\mathbf{r}', \omega_1) E_k^*(\mathbf{r}, \omega_1). \quad (12)$$

(a) Bulk, phase-mismatched (SVEA)

For plane waves $E(\omega_n, z) = A_n(z) e^{ik_n z}$, (10) with (3) as source yields

$$\frac{dA_2}{dz} = \frac{i\omega_2}{2n_2 c} \chi^{(2)} A_1^2 e^{-i\Delta k z}, \quad \Delta k = k_2 - 2k_1. \quad (13)$$

For undepleted pump and $A_2(0) = 0$,

$$A_2(z) = \frac{\omega_2 \chi^{(2)} A_1^2}{2n_2 c \Delta k} (1 - e^{-i\Delta k z}). \quad (14)$$

Substituting into (4) and projecting back into the SVEA equation at ω_1 , $dA_1/dz = (i\omega_1/(2n_1 c)) P^{\text{NL}}(\omega_1)/\varepsilon_0 \cdot e^{-ik_1 z}$, produces $dA_1/dz \supset iK_{\text{casc}} |A_1|^2 A_1$ with

$$K_{\text{casc}} = \frac{\omega_1 \omega_2}{2n_1 n_2 c^2 \Delta k} |\chi^{(2)}|^2 \times (\text{phase factor}).$$

Matching to $dA_1/dz = i(3\omega_1/(2n_1 c)) \chi_{\text{casc}}^{(3)} |A_1|^2 A_1$:

$$\boxed{\chi_{\text{casc, bulk}}^{(3)}(\omega_1) = -\frac{2\omega_1}{3c n_2 \Delta k} |\chi^{(2)}|^2 \mathcal{F}(\Delta k L),} \quad (15)$$

with $\mathcal{F} \rightarrow 1$ for $\Delta k L \gg 1$ and $\mathcal{F} \rightarrow \Delta k L/2$ for $\Delta k L \ll 1$.

(b) Single SH resonance (QNM)

Near an isolated SH quasi-normal mode,

$$G_{jl}(\mathbf{r}, \mathbf{r}'; 2\omega_1) = \frac{c^2}{2\omega_2} \frac{f_{2,j}(\mathbf{r}) f_{2,l}^*(\mathbf{r}')}{\Delta\omega + i\gamma_2/2} + (\text{bg}), \quad \Delta\omega = 2\omega_1 - \omega_2. \quad (16)$$

With $E_m(\mathbf{r}, \omega_1) = a_1 f_{1,m}(\mathbf{r})$, define

$$M \equiv \int \chi_{ijk}^{(2)}(\mathbf{r}) f_{2,i}^* f_{1,j} f_{1,k} d^3r, \quad \bar{M} \equiv \int \chi_{ijk}^{(2)}(\mathbf{r}) f_{1,i}^* f_{2,j} f_{1,k}^* d^3r = M^*. \quad (17)$$

Substituting (16) into (12) gives

$$P_i^{\text{casc}}(\mathbf{r}, \omega_1) \propto \frac{M |a_1|^2 a_1}{\Delta\omega + i\gamma_2/2} \chi_{ijk}^{(2)}(\mathbf{r}) f_{2,j}(\mathbf{r}) f_{1,k}^*(\mathbf{r}). \quad (18)$$

Projecting onto $f_{1,i}^*$ and equating to the native-Kerr projection $\int f_{1,i}^* P_i^{\text{nat}} d^3r = 3\varepsilon_0 \chi_{\text{casc}}^{(3)} |a_1|^2 a_1 \int |\mathbf{f}_1|^4 d^3r$:

$$\boxed{\chi_{\text{casc}}^{(3)} = -\frac{\varepsilon_0 \omega_1}{6} \cdot \frac{|M|^2}{(\Delta\omega + i\gamma_2/2) \int |\mathbf{f}_1|^4 d^3r}} \quad (19)$$

4 Coupled-mode equations

Projecting Maxwell's equations onto $\mathbf{f}_n^*$ (slowly-varying envelope) with phenomenological loss rates:

$$\frac{da_2}{dt} = (i\Delta\omega - \frac{\gamma_2}{2})a_2 + i\beta a_1^2 \quad (20)$$

$$\frac{da_1}{dt} = -\frac{\gamma_1}{2}a_1 + i\beta^* a_1^* a_2 + i\alpha_3 |a_1|^2 a_1 + \sqrt{\gamma_1^{(\text{rad})}} s_+ \quad (21)$$

with $s_- = -s_+ + \sqrt{\gamma_1^{(\text{rad})}} a_1$, $\Delta\omega \equiv 2\omega_1 - \omega_2$, $\gamma_1 = \gamma_1^{(\text{rad})} + \gamma_1^{(\text{nr})}$.

Manley–Rowe. In the lossless limit, differentiating $|a_n|^2$ from (20)–(21):

$$\frac{d|a_2|^2}{dt} = 2 \text{Im}[\beta a_1^{*2} a_2], \quad \frac{d|a_1|^2}{dt} = -2 \text{Im}[\beta^{(1)} a_1^{*2} a_2], \quad (22)$$

$$\frac{d}{dt}(|a_1|^2 + |a_2|^2) = 2 \text{Im}[(\beta - \beta^{(1)}) a_1^{*2} a_2], \quad (23)$$

which vanishes $\forall a_1, a_2$ iff $\beta^{(1)} = \beta$.

Modal coefficients. Substituting $\mathbf{E}(\mathbf{r}, 2\omega_1) = a_2(t)\mathbf{f}_2(\mathbf{r})$ in (10), in the slow-envelope limit:

$$\text{LHS} = [-2i\omega_2 \dot{a}_2 + 2\omega_2 \Delta\omega a_2]/\varepsilon_0,$$

$$\text{RHS} = k_2^2 a_1^2 \int \chi_{ijk}^{(2)} f_{2,i}^* f_{1,j} f_{1,k} d^3r = k_2^2 a_1^2 M.$$

Adopting the standard cavity-QED normalisation,

$$\boxed{\beta = \frac{\varepsilon_0 \omega_1}{4} \int \chi_{ijk}^{(2)}(\mathbf{r}) f_{2,i}^* f_{1,j} f_{1,k} d^3r} \quad (24)$$

$$\boxed{\alpha_3 = \frac{3\varepsilon_0 \omega_1}{8} \int \chi_{ijkl}^{(3)}(\mathbf{r}) f_{1,i}^* f_{1,j} f_{1,k} f_{1,l}^* d^3r} \quad (25)$$

5 Adiabatic elimination $\Rightarrow$ effective $\chi^{(3)}$

Assume $|\Delta\omega| \gg \gamma_2/2$, $|\dot{a}_2| \ll |\Delta\omega a_2|$. Setting $da_2/dt \rightarrow 0$:

$$a_2 = \frac{-\beta a_1^2}{\Delta\omega + i\gamma_2/2} \simeq \frac{-\beta a_1^2}{\Delta\omega} \left(1 - \frac{i\gamma_2}{2\Delta\omega}\right). \quad (26)$$

Substituting into (21):

$$\frac{da_1}{dt} = \left[-\frac{\gamma_1}{2} - \frac{|\beta|^2 \gamma_2}{2\Delta\omega^2} |a_1|^2\right] a_1 + i \left[\alpha_3 - \frac{|\beta|^2}{\Delta\omega}\right] |a_1|^2 a_1 + \sqrt{\gamma_1^{(\text{rad})}} s_+. \quad (27)$$

Effective Kerr.

$$\boxed{\alpha_3^{\text{eff}} = \alpha_3 - \frac{|\beta|^2}{\Delta\omega}} \quad (28)$$

Two-photon-like loss.

$$\gamma_3^{\text{eff}} = \frac{|\beta|^2 \gamma_2}{2 \Delta\omega^2}, \quad \gamma_3^{\text{eff}}(\Delta\omega) = \frac{|\beta|^2 (\gamma_2/2)}{\Delta\omega^2 + (\gamma_2/2)^2}. \quad (29)$$

Cascaded $\chi^{(3)}$. Inverting $\alpha_3 = (3\varepsilon_0\omega_1/8) \chi_{\text{eff}}^{(3)} \int |\mathbf{f}_1|^4 d^3r$:

$$\chi_{\text{casc}}^{(3)} = -\frac{|\beta|^2/\Delta\omega}{(3\varepsilon_0\omega_1/8) \int |\mathbf{f}_1|^4 d^3r} = -\frac{8}{3\varepsilon_0\omega_1\Delta\omega} \frac{|\beta|^2}{\int |\mathbf{f}_1|^4 d^3r}. \quad (30)$$

With $\beta = \varepsilon_0\omega_1 I/4$, $I \equiv \int \chi_{ijk}^{(2)} f_{2,i}^* f_{1,j} f_{1,k} d^3r$:

$$\boxed{\chi_{\text{casc}}^{(3)} = -\frac{\varepsilon_0 \omega_1}{6 \Delta\omega} \frac{|\int \chi_{ijk}^{(2)}(\mathbf{r}) f_{2,i}^* f_{1,j} f_{1,k} d^3r|^2}{\int |\mathbf{f}_1|^4 d^3r}}. \quad (31)$$

Explicit n -dependence. Using $|\mathbf{f}_n| \sim 1/\sqrt{\varepsilon_0 n_n^2 V_{\text{eff}}}$:

$$\int |\mathbf{f}_1|^4 d^3r \sim \frac{\zeta_1}{\varepsilon_0^2 n_1^4 V_{\text{eff}}}, \quad (32)$$

$$\left| \int \chi_{ijk}^{(2)} f_{2,i}^* f_{1,j} f_{1,k} d^3r \right|^2 \sim \frac{|\chi^{(2)}|^2 \tilde{\eta}^2}{\varepsilon_0^3 n_2^2 n_1^4 V_{\text{eff}}}. \quad (33)$$

All n_1 powers cancel:

$$\boxed{\chi_{\text{casc}}^{(3)} \sim -\frac{\omega_1}{6 n_2^2 \Delta\omega} |\chi^{(2)}|^2 \eta_{\text{geom}}, \quad \eta_{\text{geom}} \equiv \tilde{\eta}^2/\zeta_1} \quad (34)$$

Explicit form of η_{geom} . Introduce dimensionless mode profiles $u_n(\mathbf{r})$ by stripping the dimensional scale from $\mathbf{f}_n$,

$$\mathbf{f}_n(\mathbf{r}) \equiv \frac{u_n(\mathbf{r})}{\sqrt{\varepsilon_0 n_n^2 V_{\text{eff}}}}, \quad \int |u_n(\mathbf{r})|^2 d^3r = V_{\text{eff}}, \quad (35)$$

where V_{eff} is the common effective mode volume (the normalisation (9) is reproduced for non-magnetic media with $\tilde{\varepsilon}_n \sim n_n^2$ inside the mode and electric/magnetic equipartition). The u_n are

$O(1)$ where the mode is concentrated and vanish far from it. Inserting (35) into the integrals introduced for the scaling estimate,

$$\int |\mathbf{f}_1|^4 d^3r = \frac{1}{\varepsilon_0^2 n_1^4 V_{\text{eff}}^2} \int |u_1|^4 d^3r, \quad (36)$$

$$\left| \int \chi_{ijk}^{(2)} f_{2,i}^* f_{1,j} f_{1,k} d^3r \right|^2 = \frac{1}{\varepsilon_0^3 n_1^4 n_2^2 V_{\text{eff}}^3} \left| \int \chi_{ijk}^{(2)} u_{2,i}^* u_{1,j} u_{1,k} d^3r \right|^2, \quad (37)$$

fixes the dimensionless geometric factors as

$$\boxed{\zeta_1 \equiv \frac{1}{V_{\text{eff}}} \int |u_1(\mathbf{r})|^4 d^3r, \quad \tilde{\eta} \equiv \frac{1}{|\chi^{(2)}| V_{\text{eff}}} \left| \int \chi_{ijk}^{(2)}(\mathbf{r}) u_{2,i}^*(\mathbf{r}) u_{1,j}(\mathbf{r}) u_{1,k}(\mathbf{r}) d^3r \right|.} \quad (38)$$

Combining the two yields the closed-form

$$\boxed{\eta_{\text{geom}} = \frac{\tilde{\eta}^2}{\zeta_1} = \frac{\left| \int \chi_{ijk}^{(2)}(\mathbf{r}) u_{2,i}^* u_{1,j} u_{1,k} d^3r \right|^2}{|\chi^{(2)}|^2 V_{\text{eff}} \int |u_1(\mathbf{r})|^4 d^3r}.} \quad (39)$$

Equivalently, eliminating the dimensionless profiles in favour of the Brillouin-normalised $\mathbf{f}_n$,

$$\eta_{\text{geom}} = \frac{\varepsilon_0 n_2^2 \left| \int \chi_{ijk}^{(2)}(\mathbf{r}) f_{2,i}^* f_{1,j} f_{1,k} d^3r \right|^2}{|\chi^{(2)}|^2 \int |\mathbf{f}_1(\mathbf{r})|^4 d^3r}, \quad (40)$$

which back-substituted into (34) reproduces (31) exactly (the n_2^2 cancels the $1/n_2^2$ in the prefactor and the ε_0 restores the tensorial form).

Bounds and limits. By Cauchy–Schwarz on the integrand of (39), with $|\chi^{(2)}(\mathbf{r})| \leq |\chi^{(2)}|$ on its support and $\int |u_2|^2 d^3r = V_{\text{eff}}$,

$$\left| \int \chi_{ijk}^{(2)} u_{2,i}^* u_{1,j} u_{1,k} d^3r \right|^2 \leq |\chi^{(2)}|^2 V_{\text{eff}} \int |u_1|^4 d^3r \implies 0 \leq \eta_{\text{geom}} \leq 1.$$

For a uniform $\chi^{(2)}$ filling a fraction $V_{\text{NL}}/V_{\text{eff}}$ of the mode with perfectly aligned, top-hat profiles inside the nonlinear region, $\tilde{\eta}, \zeta_1 \sim V_{\text{NL}}/V_{\text{eff}}$ and hence $\eta_{\text{geom}} \sim V_{\text{NL}}/V_{\text{eff}}$ — the nonlinear fill fraction weighted by the tensor projection. Realistic (I guess) BIC/qBIC LiNbO₃ designs give $\eta_{\text{geom}} \approx 0.1\text{--}0.5$; a value $\eta_{\text{geom}} \approx 0.3$ is used in the numerical estimates (in the presentation particularly).

6 Bandwidth–enhancement sum rule

For $|a_1|^2$ modulated at Ω , $\delta a_2 = i\beta \delta(a_1^2)/(-i(\Delta\omega - \Omega) + \gamma_2/2)$. The frequency-dependent cascaded Kerr:

$$\alpha_3^{\text{eff}}(\Omega) = \frac{i|\beta|^2}{\gamma_2/2 + i(\Omega - \Delta\omega)}, \quad |\alpha_3^{\text{eff}}(\Omega)| = \frac{|\beta|^2}{\sqrt{(\gamma_2/2)^2 + (\Omega - \Delta\omega)^2}}. \quad (41)$$

$|\alpha_3^{\text{eff}}(\Omega)|^2$ is Lorentzian, FWHM γ_2 , peak $4|\beta|^4/\gamma_2^2$, area

$$\int d\Omega |\alpha_3^{\text{eff}}(\Omega)|^2 = \frac{2\pi |\beta|^4}{\gamma_2}. \quad (42)$$

At $|\Delta\omega| \sim \gamma_2$, $|\alpha_3^{\text{eff}}|_{\text{max}} \sim |\beta|^2/\gamma_2$, $\Delta\Omega \sim \gamma_2$. With $\omega_1/\gamma_2 = Q_2/2$:

$$|\chi_{\text{casc}}^{(3)}|_{\text{max}} \sim \frac{Q_2}{12 n_2^2} |\chi^{(2)}|^2 \eta_{\text{geom}}. \quad (43)$$

Multiplying by $\Delta f_{\text{usable}} \sim \gamma_2/(2\pi)$:

$$|\chi_{\text{casc}}^{(3)}|_{\text{max}} \times \Delta f_{\text{usable}} \lesssim \frac{\omega_1 |\chi^{(2)}|^2 \eta_{\text{geom}}}{12\pi n_2^2} \quad (Q_2\text{-independent}). \quad (44)$$

7 Per-photon nonlinear phase

On-resonance pump, fully-radiative limit $\gamma_1^{(\text{rad})} = \gamma_1$: $|a_1|^2 = (4/\gamma_1)|s_+|^2$, $\phi_{\text{NL}} = \alpha_3^{\text{eff}}|a_1|^2 \cdot (2/\gamma_1)$.

$$\frac{\phi_{\text{NL}}}{|s_+|^2} = \frac{8 \alpha_3^{\text{eff}}}{\gamma_1^2} \quad (45)$$

For Architecture A (native), $\alpha_3^{\text{eff,A}} = \alpha_3 \propto \chi_{\text{native}}^{(3)}$. For B (cascade),

$$\alpha_3^{\text{eff,B}} = -\frac{|\beta|^2}{\Delta\omega + i\gamma_2/2} \propto \frac{|\chi^{(2)}|^2 \eta_{\text{geom}}}{n_2^2 \Delta\omega} \propto Q_2. \quad (46)$$

Define

$$R \equiv \frac{|\alpha_3^{\text{eff,B}}|}{|\alpha_3^{\text{eff,A}}|} = \frac{|\beta|^2}{|\alpha_3| |\Delta\omega + i\gamma_2/2|} \stackrel{\Delta\omega \gg \gamma_2}{\simeq} \frac{|\chi_{\text{casc}}^{(3)}|}{|\chi_{\text{native}}^{(3)}|. \quad (47)$$

8 Kerr-driven observables

SPM. Linear in α_3^{eff} :

$$\frac{\phi_{\text{NL}}^{\text{A+B}}}{\phi_{\text{NL}}^{\text{A}}} = 1 + R. \quad (48)$$

XPM. Same scaling, with $\alpha_3^{\text{XPM}} = 2\alpha_3^{\text{SPM}}$:

$$\frac{\alpha_3^{\text{XPM, A+B}}}{\alpha_3^{\text{XPM, A}}} = 1 + R. \quad (49)$$

Bistability / switching. Steady state of (21) at detuning δ_p :

$$\left[\frac{\gamma_1}{2} - i(\delta_p + \alpha_3^{\text{eff}}|a_1|^2) \right] a_1 = \sqrt{\gamma_1^{\text{rad}}} s_+. \quad (50)$$

Scaling $y = \alpha_3^{\text{eff}}|a_1|^2/(\gamma_1/2)$, $x = \delta_p/(\gamma_1/2)$, $u = |\alpha_3^{\text{eff}}||s_+|^2 \gamma_1^{\text{rad}}/(\gamma_1/2)^3$:

$$y[1 + (x - y)^2] = u. \quad (51)$$

Three roots iff

$$\text{sign}(\delta_p) = -\text{sign}(\alpha_3^{\text{eff}}), \quad |\delta_p| > \sqrt{3} \gamma_1/2. \quad (52)$$

Cusp threshold $|a_1|_c^2 = \sqrt{3}\gamma_1/(2|\alpha_3^{\text{eff}}|)$:

$$|s_+|_{\text{bist}}^2 \propto \frac{\gamma_1^2}{|\alpha_3^{\text{eff}}|}, \quad (53)$$

$$\boxed{\frac{|s_+|_{\text{bist}}^{2, \text{A+B}}}{|s_+|_{\text{bist}}^{2, \text{A}}} = \frac{E_{\text{switch}}^{\text{A+B}}}{E_{\text{switch}}^{\text{A}}} = \frac{1}{1+R}}. \quad (54)$$

FWM. Write $a_1 = a_p + a_s e^{-i\Omega t} + a_i e^{+i\Omega t}$ with $|a_s|, |a_i| \ll |a_p|$. Collecting the $e^{+i\Omega t}$ component of $|a_1|^2 a_1$ gives source $\mathcal{S}_i = i\alpha_3^{\text{eff}} a_p^2 a_s^*$ and detuning renormalisation $2|a_p|^2 a_i$ (dropped below). Idler steady state:

$$a_i = \frac{i\alpha_3^{\text{eff}} a_p^2 a_s^*}{\gamma_1/2 - i\Omega} \simeq \frac{i\alpha_3^{\text{eff}} a_p^2 a_s^*}{\gamma_1/2} \quad (|\Omega| < \gamma_1/2). \quad (55)$$

Using $|s_-^{(i)}|^2 = \gamma_1^{\text{rad}} |a_i|^2$ and $|a_s| = 2\sqrt{\gamma_1^{\text{rad}}} |s_+^{(s)}|/\gamma_1$:

$$\eta_{\text{FWM}} \equiv \frac{|s_-^{(i)}|^2}{|s_+^{(s)}|^2} \propto \frac{|\alpha_3^{\text{eff}}|^2 |a_p|^4}{(\gamma_1/2)^2}. \quad (56)$$

$$\boxed{\frac{\eta_{\text{FWM}}^{\text{A+B}}}{\eta_{\text{FWM}}^{\text{A}}} = (1+R)^2}. \quad (57)$$

9 Imaginary part: cascaded saturable absorber

From (19), writing $1/(\Delta\omega + i\gamma_2/2) = (\Delta\omega - i\gamma_2/2)/[\Delta\omega^2 + (\gamma_2/2)^2]$:

$$\text{Im } \chi_{\text{casc}}^{(3)}(\Delta\omega) = \frac{\varepsilon_0 \omega_1}{6} \cdot \frac{|M|^2 (\gamma_2/2)}{[\Delta\omega^2 + (\gamma_2/2)^2] \int |\mathbf{f}_1|^4 d^3r}, \quad (58)$$

$$\gamma_3^{\text{eff}}(\Delta\omega) = \frac{|\beta|^2 (\gamma_2/2)}{\Delta\omega^2 + (\gamma_2/2)^2}, \quad \gamma_3^{\text{eff}}(0) = \frac{|\beta|^2}{\gamma_2/2} = \frac{2Q_2}{\omega_2} |\beta|^2. \quad (59)$$

At $\Delta\omega = 0$ (Lorentzian peak, FWHM γ_2):

$$\text{Im } \chi_{\text{casc}}^{(3)}|_{\Delta\omega=0} \sim \frac{Q_2}{6 n_2^2} |\chi^{(2)}|^2 \eta_{\text{geom}}. \quad (60)$$

10 Amplitude modulator: $\Delta\omega(t)$ transfer function

Linearise $\Delta\omega(t) = \Delta\omega_0 + \delta\Omega \cos(\Omega_{\text{mod}} t)$ in the SH equation (20). The positive-frequency sideband of δa_2 :

$$A_+(\Omega_{\text{mod}}) = \frac{(\delta\Omega/2) a_{2,0}}{\gamma_2/2 + i(\Omega_{\text{mod}} - \Delta\omega_0)}, \quad a_{2,0} = \frac{-\beta a_1^2}{\Delta\omega_0 + i\gamma_2/2}. \quad (61)$$

$$\left| \frac{\delta a_2}{a_{2,0}} \right|_{\Omega_{\text{mod}}} = \frac{\delta\Omega/2}{\sqrt{(\gamma_2/2)^2 + (\Omega_{\text{mod}} - \Delta\omega_0)^2}}. \quad (62)$$

Peak at $\Omega_{\text{mod}} = \Delta\omega_0$, DC at $\Omega_{\text{mod}} = 0$:

$$\left| \frac{\delta a_2}{a_{2,0}} \right|_{\text{peak}} = \frac{\delta\Omega}{\gamma_2}, \quad \left| \frac{\delta a_2}{a_{2,0}} \right|_{\text{DC}} = \frac{\delta\Omega}{\sqrt{\Delta\omega_0^2 + (\gamma_2/2)^2}}. \quad (63)$$

AC-peak/DC enhancement:

$$\boxed{\frac{|\delta\alpha_3^{\text{eff}}|_{\text{peak}}}{|\delta\alpha_3^{\text{eff}}|_{\text{DC}}} = \frac{\sqrt{\Delta\omega_0^2 + (\gamma_2/2)^2}}{\gamma_2} \xrightarrow{\Delta\omega_0 \gg \gamma_2} \frac{|\Delta\omega_0|}{\gamma_2}}. \quad (64)$$

11 Sensor: $d\alpha_3^{\text{casc}}/d\Delta\omega$

With $b \equiv \gamma_2/2$, $\alpha_3^{\text{casc}}(\Delta\omega) = -|\beta|^2 \Delta\omega / [\Delta\omega^2 + b^2]$:

$$\frac{d\alpha_3^{\text{casc}}}{d\Delta\omega} = -|\beta|^2 \frac{b^2 - \Delta\omega^2}{[\Delta\omega^2 + b^2]^2}. \quad (65)$$

Extrema: global max at $\Delta\omega = 0$, local max at $|\Delta\omega| = b\sqrt{3}$ (factor 8 smaller).

$$\boxed{\left| \frac{d\alpha_3^{\text{casc}}}{d\Delta\omega} \right|_{\Delta\omega=0} = \frac{|\beta|^2}{b^2} = \frac{4|\beta|^2}{\gamma_2^2} \propto Q_2^2}. \quad (66)$$

Using (45),

$$\left| \frac{d\phi_{\text{NL}}}{d\Delta\omega} \right|_{\text{max}} = \frac{32 |s_+|^2 |\beta|^2}{\gamma_1^2 \gamma_2^2} \propto Q_1^2 Q_2^2. \quad (67)$$

12 Principal result

$$\boxed{\chi_{\text{casc}}^{(3)}(\omega_1) = -\frac{\varepsilon_0 \omega_1}{6 \Delta\omega} \frac{\left| \int \chi_{ijk}^{(2)}(\mathbf{r}) f_{2,i}^* f_{1,j} f_{1,k} d^3r \right|^2}{\int |\mathbf{f}_1(\mathbf{r})|^4 d^3r} \sim -\frac{\omega_1}{6 n_2^2 \Delta\omega} |\chi^{(2)}|^2 \eta_{\text{geom}}}, \quad (68)$$

valid for $\gamma_2/2 \ll |\Delta\omega| \ll \omega_1$, $|\beta a_1| \ll |\Delta\omega|$.